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Fast Human Missions to Mars Using Direct Fusion Drive with a Nuclear Thermal Stage

Ms. Stephanie J Thomas[*,a], **Mr. Michael Paluszek**[a], **and Dr. Charles Swanson**[a]

[a]*Princeton Satellite Systems, Plainsboro, NJ*
[*]Corresponding author

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Abstract

The Direct Fusion Drive (DFD) is a nuclear fusion engine concept that produces both thrust and electric power. It employs a field reversed configuration with an odd-parity rotating magnetic field heating system to heat the plasma to fusion temperatures. The engine uses deuterium and helium-3 as fuel and hydrogen that is heated in the scrape-off layer for thrust augmentation. In this way variable exhaust velocity and thrust is obtained. The exhaust velocity for a given power is limited by the physics of the engine so that low exhaust velocities (< 50 km/s), and high thrusts, cannot be achieved. Adding a nuclear thermal engine provides an improved mix of thrust and high exhaust velocity allowing for faster and more efficient missions than either technology can achieve alone.

This paper presents the design of spacecraft for human missions to Mars. The mission is for two weeks in Mars orbit or on the surface.The mission uses NASA's Deep Space Habitat to house the crew, just to provide a payload mass. The mission uses a nuclear thermal engine to get through the Earth's radiation belts quickly. Using the nuclear thermal engine saves time and reduces the crew's exposure to radiation. The NTP also provides a high thrust abort option. The mission is almost a year for a two week stay. Longer stays allow for better return geometries and lower trip times.

The paper includes a detailed design of the Direct Fusion Drive engine. The computation of the specific power for the engine is presented along with a full mass budget for the engine. The paper provides a conceptual design for the nuclear thermal engine and its integration into the vehicle.

Keywords: fusion propulsion, nuclear thermal propulsion, mars, space power

1. Acronyms

CLR Conduction Limited Regime

CT Computer Tomography

EARS Energy and Ash Removal Shell

ENN ENN Energy Group 新奥

FRC Field Reversed Configuration

DARPA Defense Advanced Research Projects Agency

DFD Direct Fusion Drive

MeV Million electron volts 6×10^{-14} J

NERVA Nuclear Engine for Rocket Vehicle Application

NIAC NASA Innovative Advanced Concepts

NTP Nuclear Thermal Propulsion

PFRC Princeton Field-Reversed Configuration

PSS Princeton Satellite Systems

RF Radio Frequency

RMF Rotating Magnetic Field

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2. Introduction

Direct Fusion Drive (DFD) is a nuclear fusion engine concept that produces both thrust and electric power. It employs a field reversed configuration (FRC) with an odd-parity rotating magnetic field heating system to heat the plasma to fusion temperatures. Fusion energy is extracted through a novel cool plasma flow surrounding the engine core, moderating the specific impulse and increasing the thrust. Figure 1 shows an artist's conception of the engine, showing the array of magnets along the axis and two mirror magnets at the ends. The red particles represent the fusion reactants, and around it flows cool plasma extracting the fusion energy.

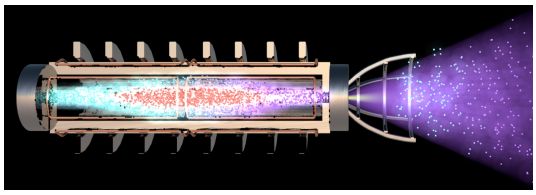


Figure 1: Artist's conception of the DFD rocket. The fusion region is red and the shell flow is cyan/purple.

Human missions to the planets have been planned since before the engine of the Apollo program in the early 1970's. At that time the most advanced propulsion option considered was nuclear fission thermal rockets which heat hydrogen flowing through a reactor core. Several of these engines were tested in the late 1960's and early 1970's, before the fission engine programs were canceled.

Recent work reports that the radiation data the Curiosity rover collected on its way to Mars found "astronauts traveling to and from Mars would be bombarded with as much radiation as they'd get from a full-body CT scan about once a week for a year." [1]. Add to that the harmful effects of muscle atrophy from long-term low-gravity, the transit speed becomes a clear priority to ensure the crew's health. Consequently, chemical or nuclear thermal rocket transfers would not be sufficient for human exploration to Mars and more distant destinations. A DFD-powered transfer stage could get astronauts to Mars in months, significantly reducing radiation exposure and atrophy effects.

To demonstrate the potential of DFD technology with a nuclear thermal stage, we developed a concept for a Mars mission that uses NASA's Deep Space Habitat to house the crew [2]. Previous work demonstrated DFD's capability for asteroid deflection [3], robotic missions [4], missions to the outer planets [5] and interstellar missions [6]. In this analysis, the DFD is augmented with a nuclear thermal engine that is used whenever high thrust is required. This eliminates the need for the long Earth departure spiral and the shorter, but still long, Mars arrival spiral. The combination of a high thrust, low specific impulsive engine with a low thrust, high specific impulse engine provides considerable mission flexibility.

The baseline vehicle is shown in Figure 2 docked with the Orion. It has six 11.5 MW engines. This provides redundancy and an abort capability in case of an engine failure. The engines provide both propulsion and electric power during the mission. The nuclear thermal engine is seen in the center.

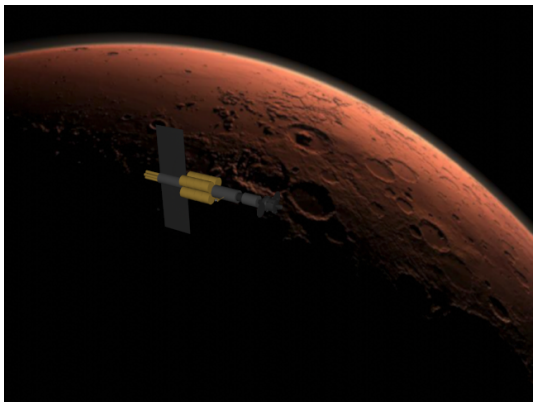


Figure 2: DFD transfer vehicle at Mars.

3. Direct Fusion Drive

This section describes the Direct Fusion Drive. The nuclear thermal rocket is described in Section 4 on Page 5, and the combined system is described in Sections 5 and 7 on page 5.

Direct Fusion Drive has been previously presented at the IAC conferences in 2012 [7, 4], 2014 [8], and 2015 [9]. At this time, all mission analysis had been performed under Princeton Satellite Systems (PSS) internal research & development funds, while work on the PFRC experiment was funded at the Princeton Plasma Physics Lab by the Department of Energy. Since then, PSS has received three research contracts from NASA to further development the DFD concept. Prior papers [10, 11, 12] describe some of the underlying physics. We will provide a summary here for the reader's convenience.

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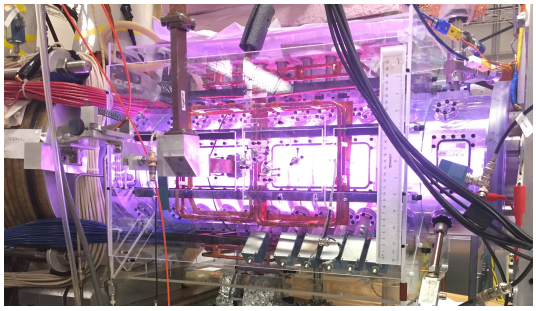


Figure 3: A plasma shot in the PFRC-2 experiment at PPPL.

3.1 DFD Summary

The Princeton Field-Reversed Configuration machine (PFRC) employs a unique radio frequency (RF) plasma heating method. Odd-parity heating was first theorized in 2000 [10] and demonstrated in the 4 cm radius PFRC-1 experiment in 2006 [11]. Experiments are ongoing with the second-generation machine, PFRC-2, which has a coil inner radius of 8 cm (Figure 3). Results from experimental studies of electron heating in PFRC-2 have surpassed theoretical predictions, with minority electron populations reaching 500 eV with pulse lengths of 300 ms. Experiments to measure ion heating with input power up to 200 kW are ongoing. When scaled up to achieve fusion parameters, PFRC would result in a 4-8 m long, 1.5 m diameter reactor producing 1 to 10 MW. This reactor would be uniquely small and clean among all fusion reactor concepts, producing very low levels of damaging neutrons. A diagram of the engine is shown in Figure 4. The startup system, for energizing the coils and starting the fusion reaction, is in the upper right corner [13]. The fusion vacuum vessel is in orange. The RMF system is depicted above the vacuum vessel. Coil refrigeration is on the right. The heat recycling system is on the bottom.

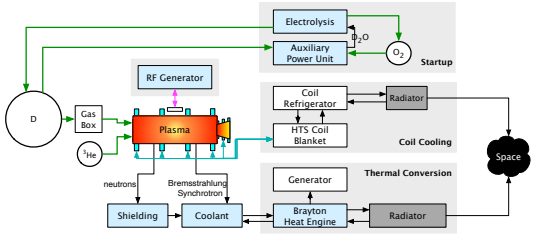


Figure 4: Block diagram of the DFD engine.

The plasma heating method uses a unique configuration of the radio frequency antenna. Attempts have been made to heat FRC plasmas with RF before, but generally with picture-frame antenna that resulted in a near-FRC plasma but with open magnetic field lines. We call this even-parity heating due to the symmetry of the induced magnetic field. Open field lines give the plasma an opportunity to escape and reduce confinement time. In contrast, each of the four PFRC antenna sections are two joined rectangles. Two pairs operate 90 degrees out of phase on adjacent sides of the plasma. An antenna (wrapped in orange Kapton tape) is clearly visible on the side of PFRC-2 in Figure 3. This results in so-called odd-parity heating — the magnetic field on one side of each figure-8 is in the opposite direction as the other side — and closed field lines in the generated FRC. Closed field lines keep the plasma trapped as it is heated. The oscillation of the currents in the RF antenna result in a rotating magnetic field, RMF, with about 0.1-1% of the strength of the axial magnetic field.

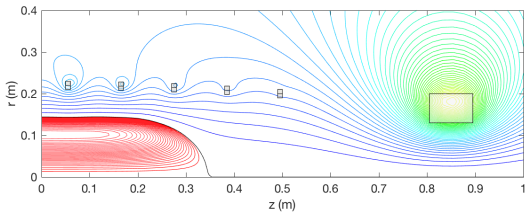


Figure 5: Closed field-lines as predicted in the PFRC core. Cool plasma flows along the open field-lines to extract the fusion energy and ash.

The PFRC is specifically designed to have ultra-low radioactivity. Being a "high-β" machine, that maintains a high β, the ratio of magnetic field to plasma pressure, it can achieve the high temperates necessary to burn helium-3 (^{3}He) and deuterium (D). These are considered "advanced fuels". This primary reaction results in no damaging neutrons, although some will be generated from D-D side reactions. The reaction equations are:

$$\mathrm{D} + {}^3\mathrm{He} \rightarrow {}^4\mathrm{He}\ (3.6\ \mathrm{MeV}) + \mathrm{p}\ (14.7\ \mathrm{MeV}) \quad (1)$$

$$\mathrm{D} + \mathrm{D} \rightarrow \mathrm{T}\ (1.01\ \mathrm{MeV}) + \mathrm{H}\ (3.02\ \mathrm{MeV}) \quad (2)$$

$$\mathrm{D} + \mathrm{D} \rightarrow {}^3\mathrm{He}\ (0.82\ \mathrm{MeV}) + \mathrm{n}\ (2.45\ \mathrm{MeV}) \quad (3)$$

$$\mathrm{D} + \mathrm{T} \rightarrow {}^4\mathrm{He}\ (3.5\ \mathrm{MeV}) + \mathrm{n}\ (14.1\ \mathrm{MeV}) \quad (4)$$

Simply burning ^{3}He-D at high temperatures in a large FRC reactor could actually result in an increase in neutrons due to the above side reactions. PFRC has three features which greatly reduce the number and impact of the neutrons from this baseline: one, the small size of the reactor ($\sim$25 cm plasma radius) results in a favorable ratio of surface area to plasma volume, reducing the wall load compared to larger machines; two, we can adjust the operating fuel ratio of ^{3}He:D upwards to as much as 3:1, sacrificing some power density for lower neutron production; three, the reactor is designed to rapidly eliminate the tritium produced by the D-D side reactions, preventing any D-T reactions from occurring. This means that the only neutrons produced are those with energy of 2.45 MeV.

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The tritium is eliminated due to its interaction with the Energy and Ash Removal Shell (EARS) surrounding the fusion region. The size of the reactor is such that the so-called FRC s-parameter,[14] which scales with the ratio of the fusion-product triton gyro-radius to the scrape-off-layer radius, is low - about 2.3. This forces the tritons to pass through the EARS repeatedly. When the tritons pass through the cool EARS plasma, electron drag causes energy to be transferred from the tritons to the EARS electrons. The triton is quickly captured by the SOL field lines and flows out the open end of the reactor. The burn-up time for energetic tritons to fuse is about 20 seconds, while the time in which it will cool and be trapped in the EARS is less than 0.01 s.[9] The same process occurs for the other fusion ash products, which are all effectively exhausted.

The PFRC is perfectly suited for use as space propulsion for two reasons: one, the configuration and small size results in a tremendous reduction of neutron production compared to other D-^{3}He approaches, as described above; two, the directed axial flow of cool EARS plasma which absorbs the energy of the fusion products. When the divertor end of the PFRC is configured with a magnetic nozzle, the reactor becomes the Direct Fusion Drive (DFD). Adding additional propellant to the external flow results in a moderate thrust, high specific impulse exhaust stream. This process is called *thrust augmentation*. The low neutron production is crucial to minimizing the shielding required while in transit, maximizing the engine specific power. The maximum thrust for a given power, based on UEDGE [15] is given in Figure 6. This leads to a practical lower exhaust velocity limit of 70 km/s assuming a 0.4 power to thrust conversion efficiency.

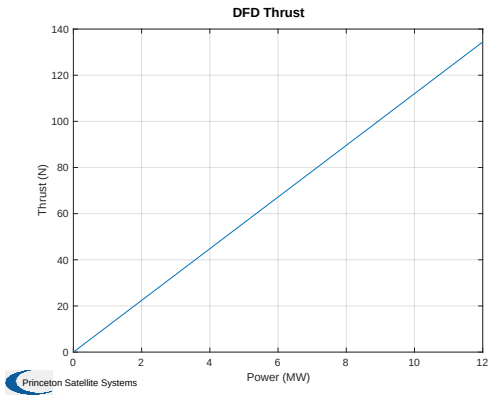


Figure 6: Maximum DFD thrust as a function of power.

The fact that DFD is so compact, has a linear form, and will be limited to 1 to about 10 MW, makes it well-suited to a variety of space missions.

3.2 DFD Power and Mass Breakdowns

Figure 7 shows a power model for a 10.42 MW$_f$ DFD engine. The plasma length is 5.2 m. 7.48 MW exits the engine in the plume. 0.1 MW is available in electricity after powering the RMF heating. Bremsstrahlung and synchrotron radiation are loss mechanisms from the plasma electrons; the RMF subsystem must compensate for this loss while heating the ions. Fortunately, modern switching RF systems are close to 100% efficient, so only a small portion of the RMF power is lost to heat before entering the plasma. This diagram assumes an electrical conversion efficiency of 60%, such as may be achieved using a Brayton cycle with a recuperator.

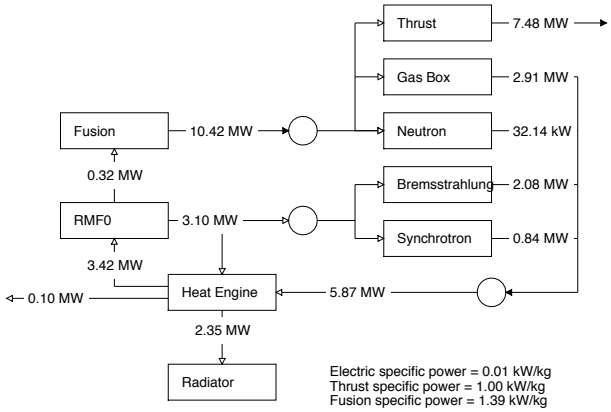


Figure 7: Power model for the DFD engine.

Figure 8 shows a mass breakdown of a 10.42 MW$_f$ engine. The specific mass computation includes not only the magnets and their cooling equipment, structure, heating and conversion subsystems, but also the shielding and radiators. The heating and conversion systems are scaled from simple specific mass factors. The mass is dominated by the radiators. 100 kW is available for electric power use in the spacecraft.

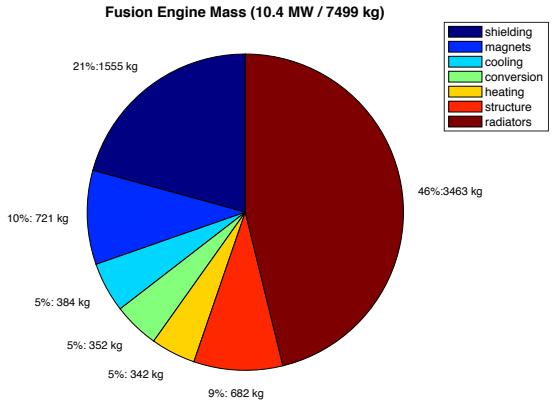


Figure 8: Mass Breakdown for the DFD engine.

The magnet subsystem includes an array of eight

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flux-conserving, superconducting magnets and two stronger mirror magnets. The magnet cryocoolers are a substantial additional mass. The axial magnets will maintain a central field of 5.3 T while the mirror magnets will require a field of about 15 to 20 T. The total engine mass, including radiators, is 7500 kg.

4. Nuclear Thermal Rocket Engines

Nuclear thermal engines were developed in the 1960's to provide higher performance upper stages for Apollo lunar missions and human Mars missions [16]. Table 1 gives the performance for the Small Engine [17]. This engine was proposed for a Space Shuttle compatible upper stage. McDonnell Douglas designed a reusable upper stage around this engine.

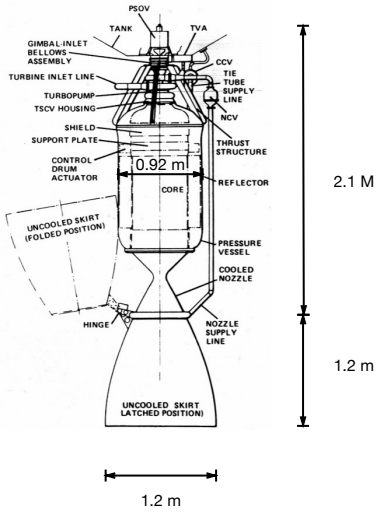


Figure 9: NASA Small NTP [18]. This is shown for informational purposes, a larger NTP is required for the thrust level required for Earth departure.

The small engine is shown in Figure 9. As shown in the table, the thrust is about 7300 N, the specific impulse is about 875 s, and the mass is 2550 kg.

Table 1: Small Nuclear Thermal Engine [18]

Parameter	Value
Chamber Pressure	310 N/cm^2
Chamber Flow Rate	8.51 kg/s
Chamber Temperature	2696 deg-K
Reactor Power	367 MW
Nozzle Flow Fraction	44.9%
Turbine Bypass Flow Fraction	11.6%
Turbopump Speed	4917 rad/s
Turbopump Shaft Work	0,93 MW
Pump Efficiency	65%
Turbine Efficiency	80%
Nozzle Valve Area	2.63 cm^2
Turbine Control Valve Area	3.02 cm^2
Length Engine	3.1 m
Length Nozzle	1.2 m
Reactor Radius	0.92 m
Nozzle Radius	1.2 m
Specific Impulse (Velocity)	8580 m/s
Thrust	7297 N
Mass	2550 kg
Thrust/Mass	2.86 N/kg

5. Mars Transport

The Mars transport is shown in Figure 10. The Small NTP engine is located between the DFD engines. The shielding in the DFD engines, and the fuel tanks, should reduce the need for NTP shielding.

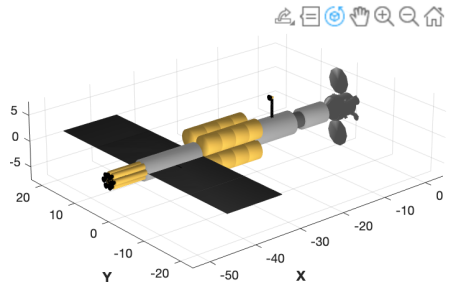


Figure 10: DFD transport with a Small NTP engine for high thrust maneuvers..

The large wings are the radiators. The solar panels on the Orion spacecraft have since been redesigned. The DFD has 6 engines which 3 of the solenoid dipoles facing fore and 3 aft. This cancels the magnetic dipole so that large torque are not produced while near the Earth. An additional coil compensates for the residual dipole [19]. A laser communication system is on the tower.

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6. Propulsion System Comparison

The maximum velocity change that an engine can achieve is limited by the structural fraction and the exhaust velocity. The fuel required for a given velocity change is

$$\gamma = e^{\frac{\Delta u}{u_e}} \tag{5}$$

$$\frac{m_f}{m_p} = \frac{1 + f(1-\gamma)}{\gamma - 1} \tag{6}$$

where f is the structural fraction. As $m_f \to \infty$

$$\frac{\Delta u}{u_e} = \log\left(\frac{1+f}{f}\right) \tag{7}$$

This is the maximum possible velocity change. Figure 11 compares NTP with DFD.

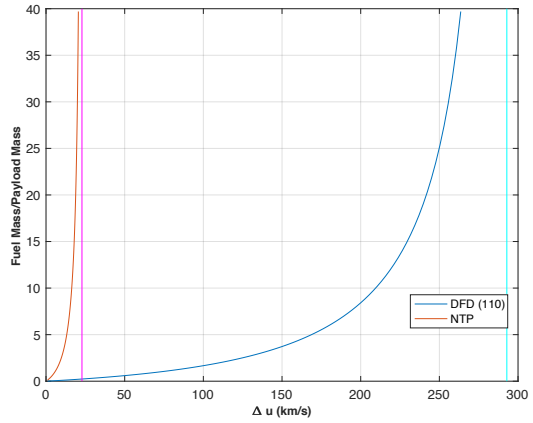


Figure 11: Fuel mass versus velocity change for DFD and NTP. The vertical lines are the limits.

An all DFD spiral from ISS altitude is shown in Figure 12. Over four weeks are required and the spacecraft spends considerable time in the radiation betls.

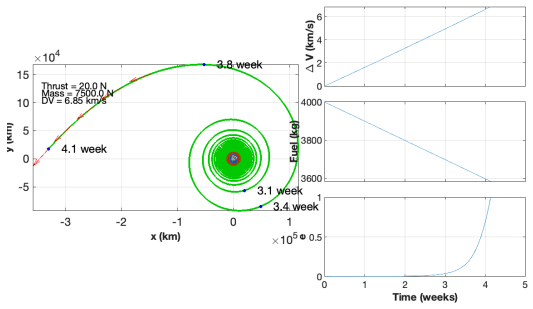


Figure 12: Earth outward spiral from an ISS orbit. This is shown for comparison purposes.

7. DFD/NTP Mars Mission

Table 2 shows the hybrid mission. The mission has a two week stay at Mars. The mission is pieced together from a series of maneuvers. It starts with an NTP Hohmann transfer through the Earth radiation belts. The remainder of the mission is done with the DFD engine, until the final return to LEO. The exhaust velocity is varied across the mission. This was done iteratively by hand. Note that constant thrust for the Earth to Mars and Mars to Earth segments is not necessarily optimal. Constant acceleration has been shown to be near optimal for straight line trajectory approximations [20]. The spirals are to the planet relative eccentricity of one and the interfaces between the spiral segments and the interplanetary segments are not continuous. The Hohmann burns require at least 100 kN of thrust for the departure and 40 kN for the return, which would require a larger engine than the small NTP shown in this paper.

Table 2: Hybrid Mission

Stage	Velocity change (km/s)	Exhaust Velocity (km/s)	Time
NTP Hohmann	1.96	8.8	1.29 (hours)
DFD Earth Outward Spiral	5.67	120.0	19.32 (days)
Earth/Mars Trajectory	36.61	110.0	90.00 (days)
DFD Mars Inward Spiral	3.27	122.0	7.70 (days)
DFD Mars Outward Spiral	3.27	125.0	7.70 (days)
Mars/Earth Trajectory	50.00	260.0	220.00 (days)
DFD Earth Inward Spiral	5.67	200.0	19.32 (days)
NTP Hohmann	1.96	8.8	1.29 (hours)
Mission	108.43		378.15 (days)

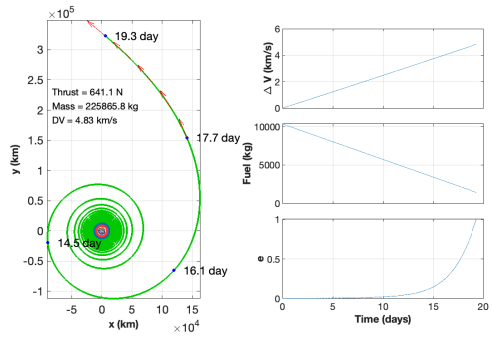


Figure 13: Earth outward spiral from above the radiation belts.

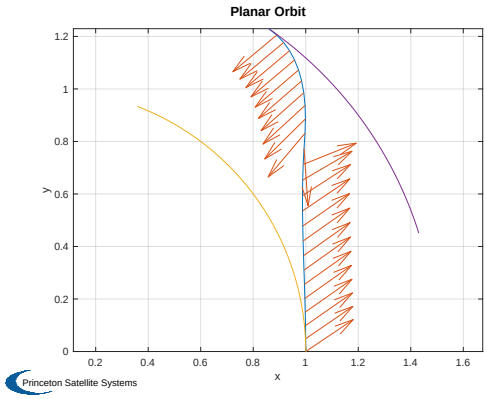


Figure 14: Earth/Mars transfer.

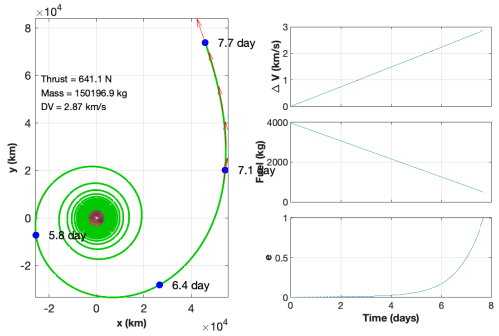


Figure 15: Mars spiral.

The return is shown in Figure 16 and in Figure 17. It is advantageous to go inside the Earth's orbit to catch up with the Earth.

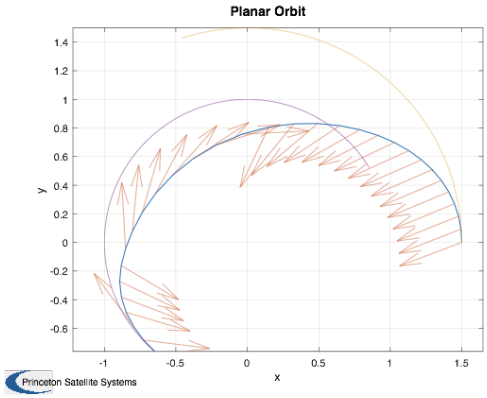


Figure 16: Mars/Earth transfer.

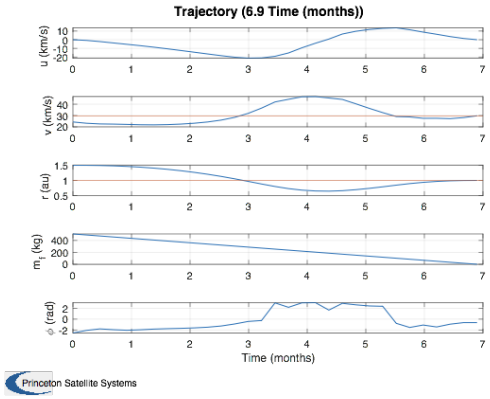


Figure 17: Mars/Earth transfer.states.

8. NTP and DFD Development Plans

PFRC, the reactor technology in DFD, is currently under development under an ARPA-E grant. ARPA-E interest is for both base load and portable power applications. It was previously funded by a NASA NIAC grant. The baseline PFRC/DFD development plan is shown in Figure 18.

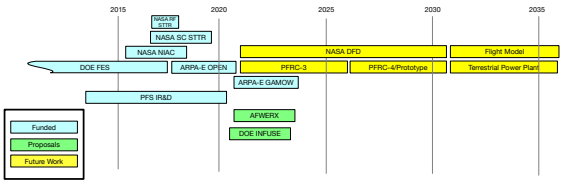


Figure 18: PFRC/DFD development plan.

This plan assumes that PFRC-3 is funded in 2021. PFRC-3 would reach fusion relevant temperatures and pressures. The plan could be accelerated in many ways. Increasing the number of sites working on PFRC technology would accelerate development. FRCs are being studied by Helion and TAE but their proposed machines use different heating mechanisms. ENN Energy in China is also working on a PFRC-like machine. Our approach, shown in the figure, is to work on the balance of plant in parallel. The ARPA-E GAMOW program does just this by focusing on critical technologies such as power electronics for fusion heating and plasma control

Both NASA and DARPA [21] plan to test a nuclear propulsion system by 2024. While nuclear rockets were tested in the 1960's and early 1970's no integrated nuclear thermal engines exist today. There were many unresolved technical issues in 1973. The nuclear rocket program was initiated in 1955 and terminated in 1973. The final down firing demonstration was in March 1969, the XE-prime which was operated at 1100 MW. The Nuclear Flight Systems Definition (NFSD) Phase II commenced in October 1969. Prior to termination, it ap-

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pears that NASA was hoping to start flying the NERVA stage in 1980 [22]. Thus a 11-13 year development time would not be unreasonable for a nuclear upper stage. Borowski describes a 13 year timeline for a small NTP engine in [17]. Thus the NTP development time frame is compatible with the DFD timeframe.

9. Conclusions

DFD has the potential to revolutionize Mars exploration and solar system exploration. Combining DFD with a nuclear thermal stage reduces trip time and astronaut exposure to radiation. Future work in optimizing the entire trajectory is required. Planetary departures and arrivals needed to be fitted together. Further work on the balance of plant for the engine is needed with the goal of improving the specific power. The heat engine, radiators and shielding are areas where mass savings may be achieved.

The mission plan can be improved. Varying the thrust and exhaust velocity per segment is expected to lead to shorter trip times. The use of the NTP should also be investigated from Mars entry.

Ongoing experiments at PPPL are working to demonstrate ion heating using odd-parity RMF. The next step in DFD development will be a fully-superconducting testbed which can heat ions to fusion-relevant temperatures of 5 to 10 keV, followed by a full-sized high-temperature fusion demonstration. At this point, a space prototype with the full balance of plant would be built, followed by the first DFD mission. Due to the simple geometry of DFD, we estimate that each testbed cycle might require 3 to 5 years. This could allow the first DFD mission to launch in 15 years or sooner.

Acknowledgements

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